



# TLChoiceNet: Quantitatively Choosing a Transfer Learning Dataset



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## SUMMARY

### PROBLEM STATEMENT

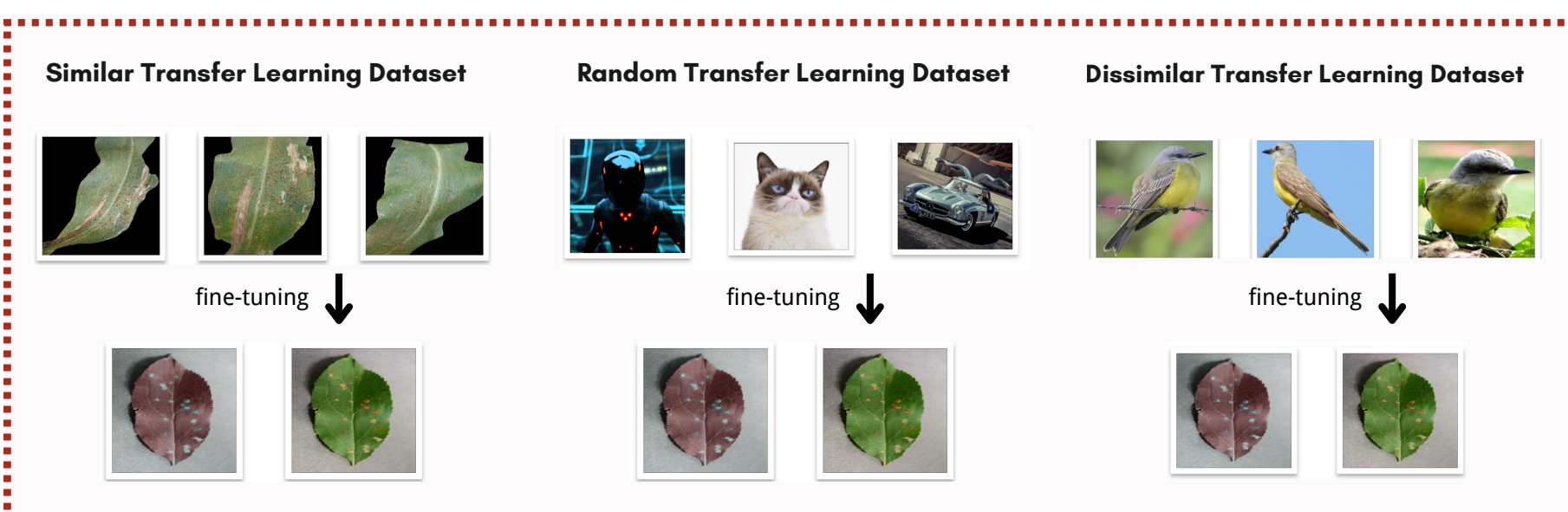
Training a performant model is substantially aided with transfer learning, primarily by preconditioning one's model upon information inside the transfer learned dataset. The questions we strive to answer are:

- If no pre-trained models available: which dataset to pre-train upon?
- Otherwise: which pre-trained model should be chosen?

### SOLUTION

- Constructed a **quantitative model called TLChoiceNet** that predicts test-set accuracy given a fine-tuning dataset and transfer learning weights.
  - Created a dataset of (dataset, model, accuracy) tuples to train and test.
  - Contrived embeddings for datasets and models as TLChoiceNet input.
- Designed a **mathematical metric** to measure similarity between two datasets and compared it with centered kernel alignment.
- Measured how model weights shift across fine-tuning, comparing if the transfer learning dataset was similar or dissimilar to the fine-tuning dataset.

## DATASET



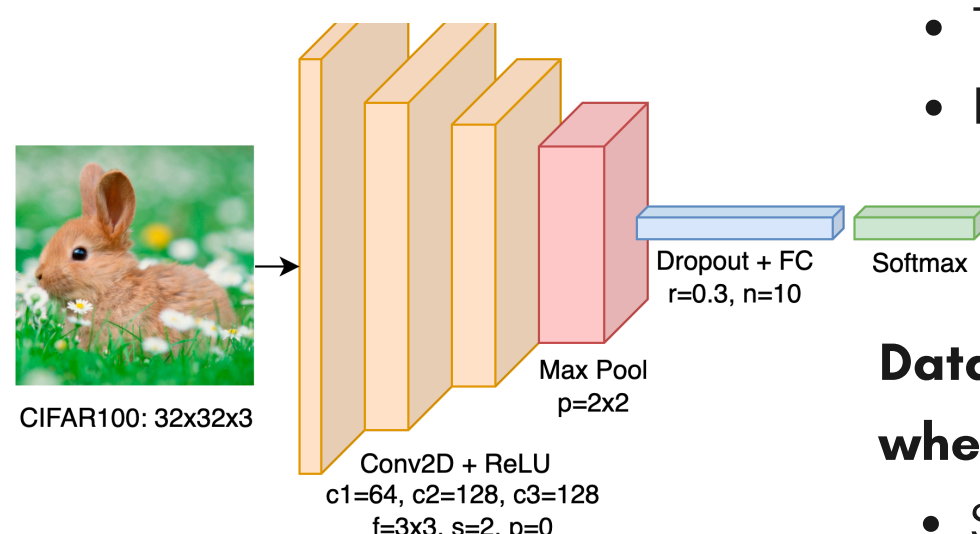
### Created dataset of 3-tuples:

- Transfer learning (TL) dataset
- Transfer learning model's weights
- Fine-tuned model's test accuracy

### Dataset consists of 40+ datapoints, where 10-class TL datasets were:

- Similar to the fine-tuning dataset
- Dissimilar from the dataset
- Random subsets of CIFAR 100

### Transfer Learning Model



## METHODOLOGY

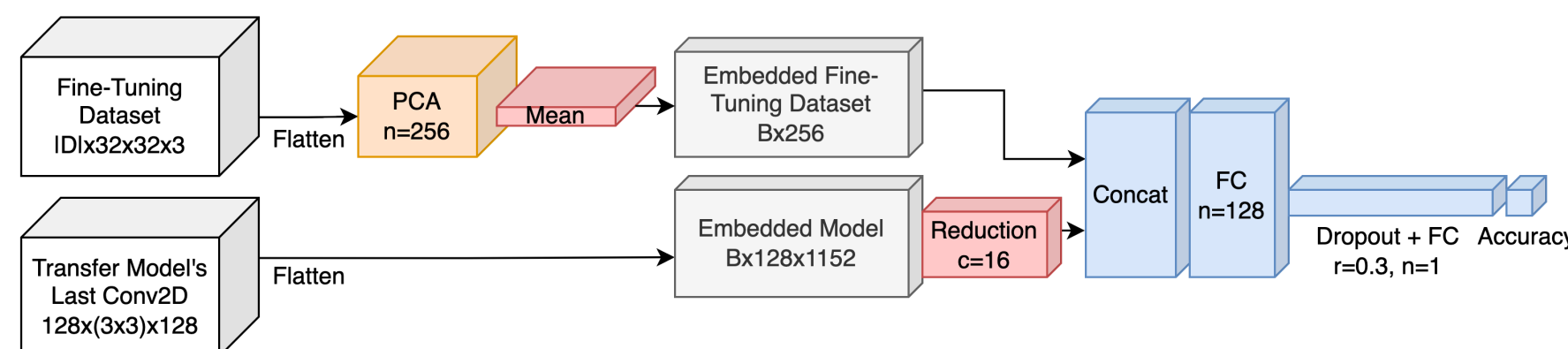
### TLCHOICENET ARCHITECTURE

#### Preprocessing of TLDS:

- Embed fine-tuning dataset using PCA and averaging across examples
- Embed TL model simply by extracting final Conv2D layer's weights

#### TLChoiceNet v1:

- LoRA-similar model reduction
- Two fully-connected layers
- Regression output: predicted test accuracy if used as TL dataset



### MEASUREMENT OF DATASET DISTRIBUTION

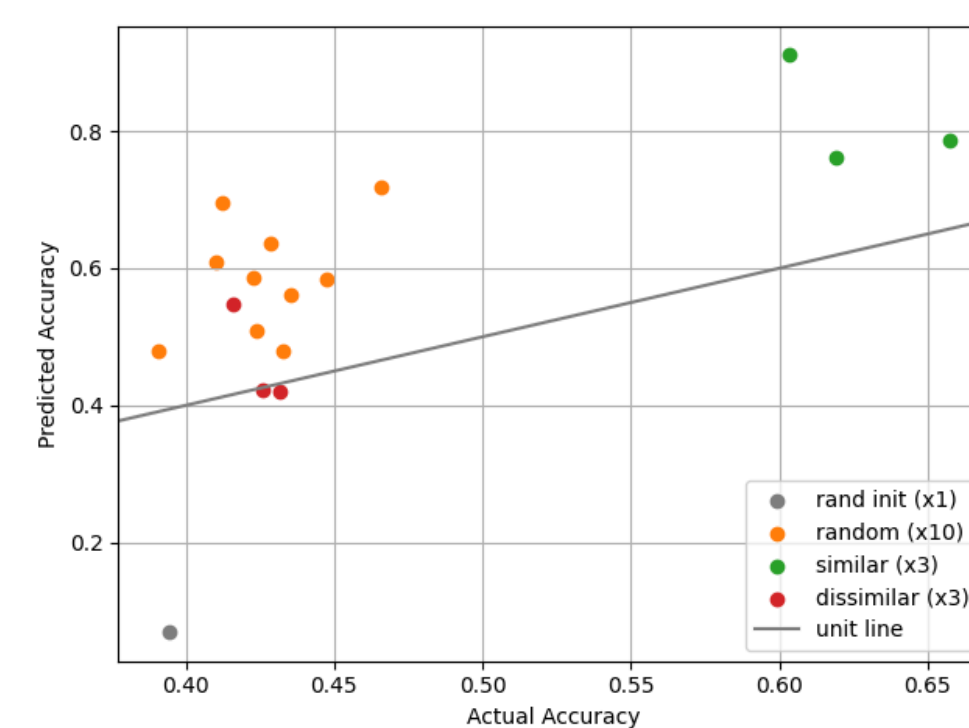
- Designed a **distance metric M** which uses **mean, skew, and kurtosis** to analyses variability of distribution between TL dataset and fine-tune dataset
1. Calculate the mean, skew, and kurtosis of each dataset's normalized pixel values. Note that we use the absolute value of skew and kurtosis to become invariant to sidedness.
  2. Fit the metric M with fine-tune accuracy to analyze the correlation.

$$M(i, j) = \sqrt{(\mu_i - \mu_j)^2 + \left( \left| \frac{1}{\sigma_i^3} \frac{1}{n} \sum_{k=1}^n (x_k - \mu_i)^3 \right| - \left| \frac{1}{\sigma_j^3} \frac{1}{m} \sum_{k=1}^m (x_k - \mu_j)^3 \right| \right)^2 + \left( \left| \frac{1}{\sigma_i^4} \frac{1}{n} \sum_{k=1}^n (x_k - \mu_i)^4 \right| - \left| \frac{1}{\sigma_j^4} \frac{1}{m} \sum_{k=1}^m (x_k - \mu_j)^4 \right| \right)^2}$$

where  $i, j$  are datasets with mean  $\mu_i, \mu_j$ , standard deviation  $\sigma_i, \sigma_j$ , and number of samples  $n, m$

## RESULTS

### TDLCHOICENET PREDICTIONS

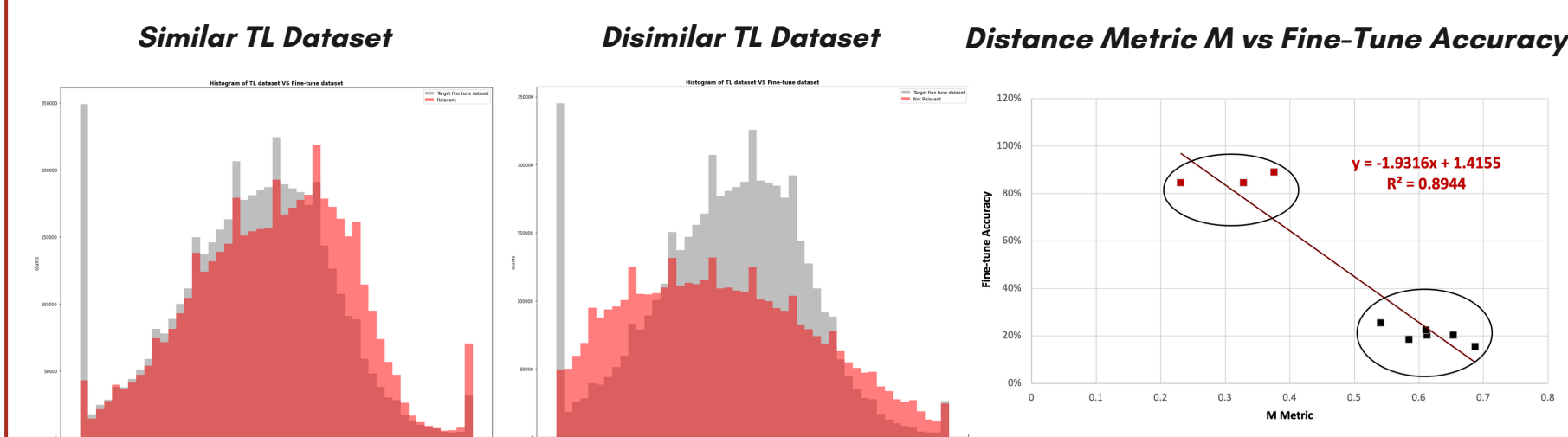


- ChoiceNet was able to predict accurately with a **MSE of 0.139** for TL network weights with an unseen fine-tune dataset.
- However, for both unseen TL weights and fine-tune data, the MSE is much higher (about 0.46) and the predicted accuracy of 10% is far off from the actual fine-tuning accuracy of 90%.

## FINDINGS

### 1. DATASET DISTRIBUTION- LOW LEVEL STATISTICS

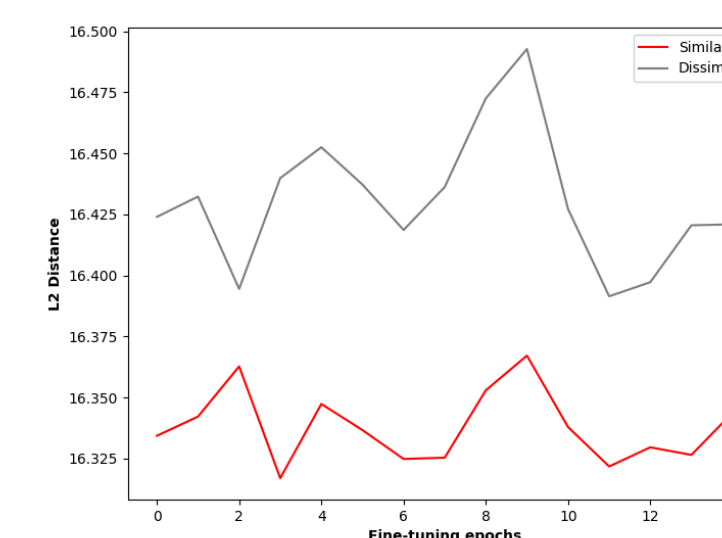
- Test fine-tune performance is correlated with the distribution of normalized pixel values between transfer and fine-tune datasets.
- Shown in the histogram below, there exists a distribution shift between dissimilar TL dataset classes and similar TL dataset classes.
- Fitting the distance metric M with test fine-tune accuracy, it can explain the variation in **fine-tune accuracy with a R2 of 0.894**.



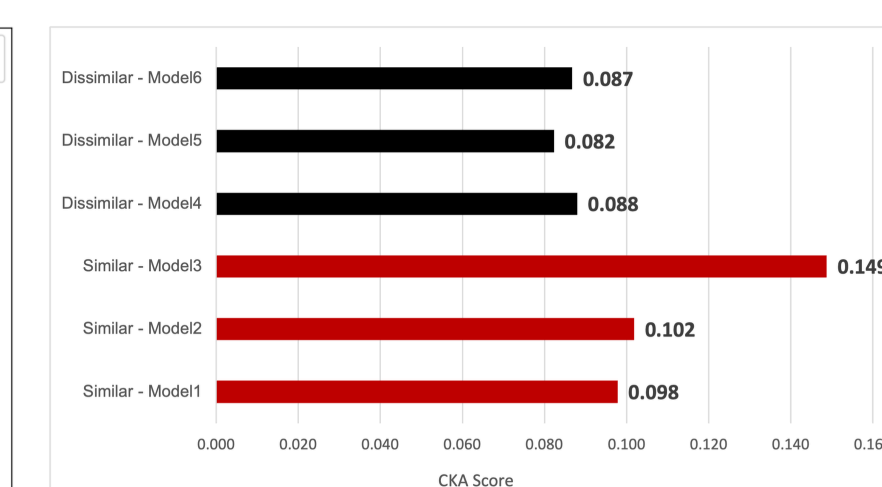
### 2. PRE-TRAINED WEIGHTS DURING FINE-TUNING

- To understand how pre-trained weight change for similar and dissimilar classes during fine-tuning, we measure L2 norm between pre-trained weights and fine-tune weights across fine-tuning epochs.
- As epochs progress, the L2 distance between the initial transfer weights and fine-tune weights increase. For dissimilar transfer weights, the L2 distance increases more than similar transfer weights.

#### L2 Distance of Weights Across Epochs



#### Centered Kernel Alignment



### 3. CENTERED KERNAL ALIGNMENT

- CKA is a similarity metric that measures the multivariate similarity of features between two neural networks.
- We found CKA score, by comparing the post-transfer learning model with the post-fine-tuning model, is higher when transfer learning with a dataset similar to the fine-tuning dataset.